5DATA006C.1 Data Visualisation and Communication.

COURSEWORK

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Introduction

London Fire Brigade (LFB) is reputed to be one of the busiest fire and rescue services in the world. This report is a visual, data-driven narrative of how demand in the services provided by the brigade is changing over time, per incident type, geography and by property type, and the urgency of the first pump arrival using incident records in 2018 to 2024. It is not only to describe a series of six separate charts but to relate them into a logical story: when and where London requires the fire service most, what types of emergencies fill the workload and the performance differentials around the entire city.

Part A

Data Storytelling

How do incident type, location and property risk influence London Fire Brigade demand and response times between 2018 and 2024?

Between 2018 and 2024, The crews of London Fire Brigade had to address approximately one million incidents, which emphasized the high pressure on one of the busiest fire services in the world. This discussion looks at the motivators of such demand, investigation of types of incidences, geographical hot spots, high risk situations and how emergency response is delivered promptly to those most in need. (Greater London Authority, 2025)

Count of incidents

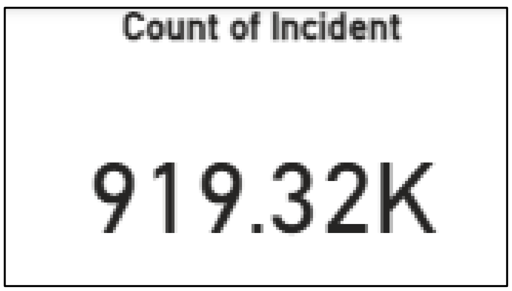

Figure 1(kpi card)

The story begins with a single figure. One of the key performance indicators cards displays the number of incidents that have been recorded in all of London in the period. The fact is that on the surface, it is obvious that the demand for the services that the brigade provides is continuously high. The totals are impressive even in the case of individual years filtration. This puts the LFB planners and policymakers directly in the situation, it is not that the brigade is struggling with spikes in the activity, but long-term, sustained pressure. Any alterations in the pattern of incidents, consequently, have grave consequences on staffing, financial capacity and response capacity.

Count of IncidentNumber by YearMonth and IncidentGroup

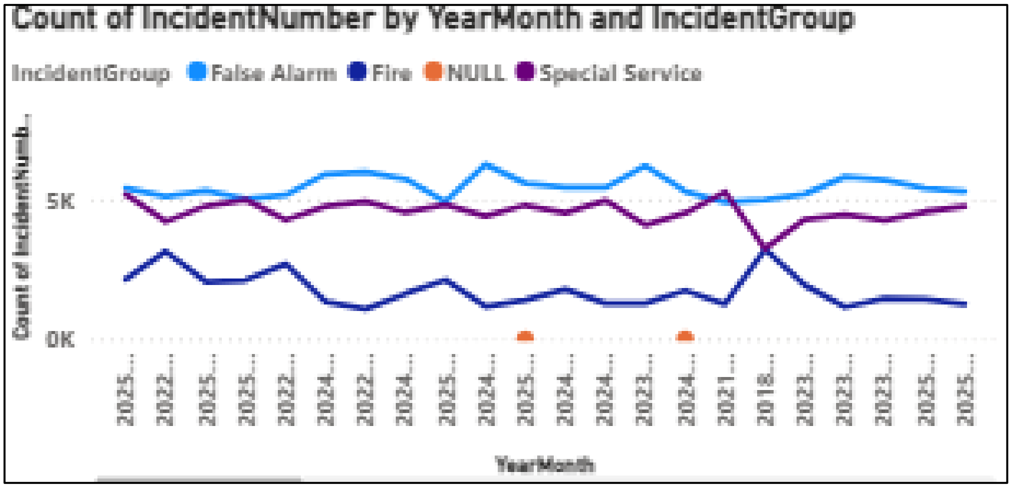

Figure 2 (Line chart)

On going beyond the headline, a line chart of monthly incidents by incident group shows the distribution of this demand over time. The most visible line over the seven years is the false alarm, which wins most of the vacancies in comparison to fires and special services. However, seasonal variations can be observed, but the overall situation has not changed: the false alarm cases occupy a significant portion of the work of the brigade each year. Fire incidents follow behind false alarms but are still significant whereas special service calls, including rescues and road traffic collisions, compose a minor but consistent part of the activity. This time-warped perspective indicates that although the risk scenario in London has been fluctuating monthly, the general framework of demand is still very steady.

Count of IncidentNumber by IncidentGroup

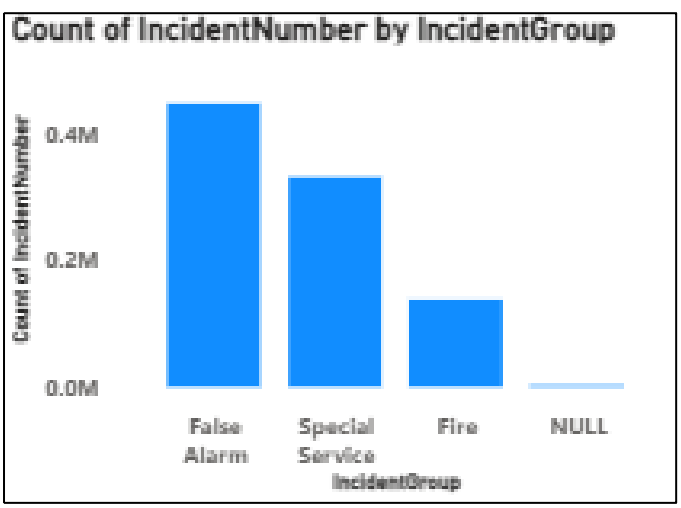

Figure 3(column chart)

A better picture of that structure is the clustered column chart that summaries the incident by the group. In this case, the prevalence of false alarms will become clear. Alarms cause more calls, but which do not lead to fires than fires themselves. This has obvious operational implications. Each false alarm also needs crews, vehicles and time mobilised and this may be at the expense of other life-critical incidents. Simultaneously, the fact that the number of special service calls is large shows the way the modern fire service developed. The LFB is not just responding to fire any longer, but to a vast array of emergencies which indicate the intricacy of life in a global city. This demand heterogeneity involves demand flexibility in training, equipment and planning.

Count of IncidentNumber. First IncidentGroup and Count of CalYear by InGeo_BoroughName.

Figure 4(geography map)

But it is not all demand that is spread evenly throughout London. Geographical variation is drastic on a map of incidents by borough. The densely populated and central boroughs are represented by bigger bubbles which imply the vast number of calls compared to many external territories. These spatial patterns would still be apparent at the time of filtering based on year or incident type. The map narrates a well-known and significant tale: where people, businesses and infrastructure are the most densely located, incidents occur more often. This supports decision-makers the need to match the location of stations, crew deployment and prevention activities with local levels of risk other than implementing a one-size-fits-all strategy across the city.

Average First Pump Response Time – Top 10

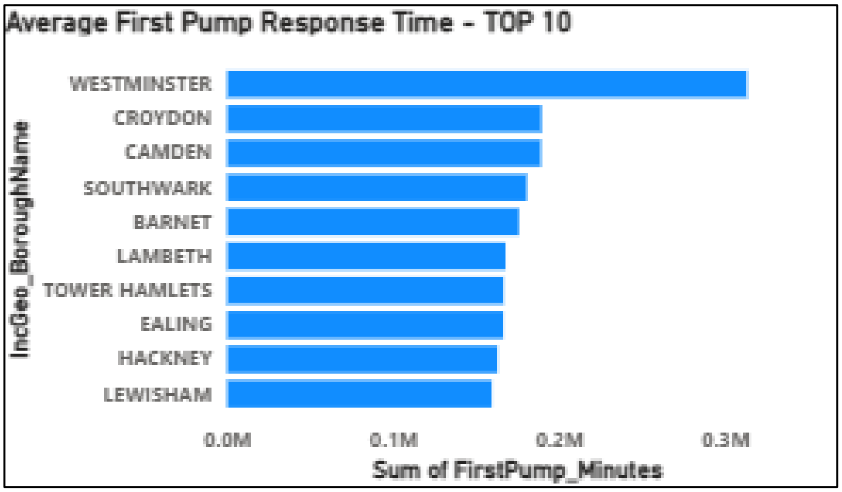

Figure 5(bar chart)

The main question posed by high demand is the rate of brigade response. Mean first-pump response time in the ten busiest boroughs is plotted in a bar chart that causes a shift in the focus of the call volume toward performance. Though the differences are not significant, they are still considerable. Certain high-need boroughs record significantly quicker responses as compared to some others. These gaps could be due to factors like congestion, network coverage and efficiency of call-handling of the stations. It is important to concentrate on the boroughs that are busiest, which will enable us to make the best contribution in terms of the small improvements in to improving the response times to benefit the safety of the population.

Count of IncidentNumber by PropertyCategory and Property Type.

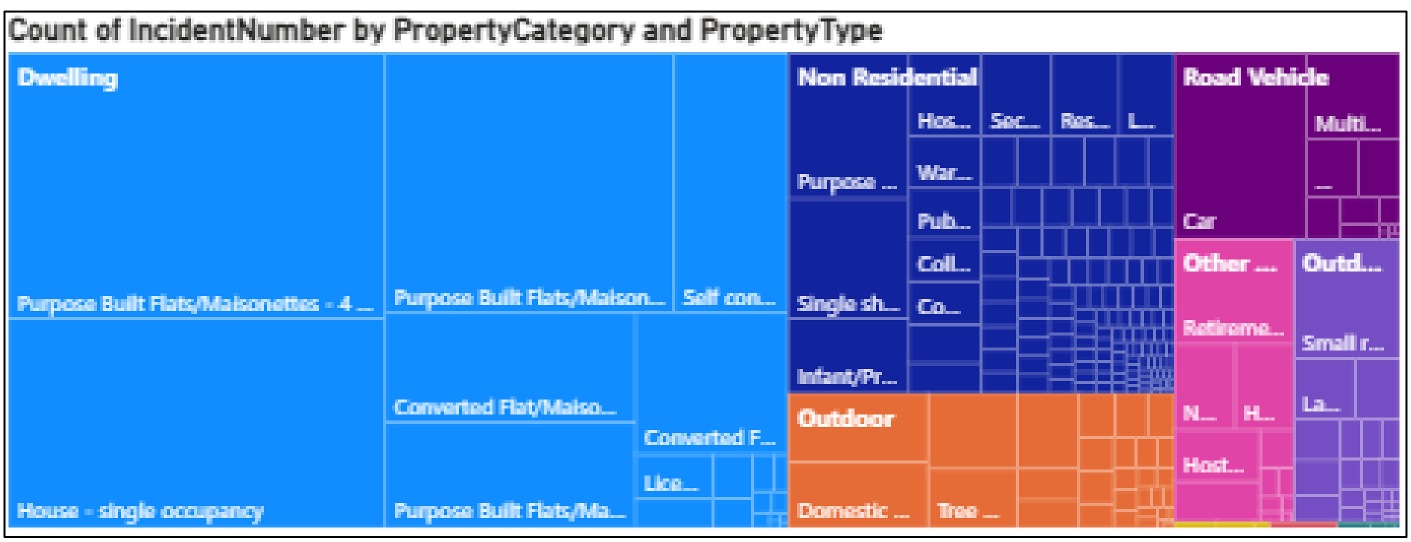

Figure 6(tree map)

The last image changes the emphasis on the location to the possessions. A treemap of incidents by property type demonstrates that dwellings predominate emphasizing the fact that many incidents of LFB are found in a residential environment, especially in a flat or a house which underscores the role of home fire safety. There are also non-residential facilities like hospitals and schools, which are also emphasized since crucial facilities should be protected. The large range of operations is shown in the cases with road vehicles and outdoor locations, and the need for constraints is seen in smaller cases of unknown.

Combined, the six images demonstrate the fire service of London as being under regular pressure, conditioned by false alarms, residential risk, uneven demand in the boroughs, and subtle response performance, demonstrating challenges of everyday operations and human struggles.

Part B

a) Methodology

In this coursework I choose the data sets about the London fire brigade incident on data.gov.uk since they are rich datasets, well-structured and directly linked to the issue of safety of the people. Two Excel files were given, one of them being 2018-2023 incidents and another 2024 onwards incidents. I brought them into Power BI and examined the structure and ensured that they all had the same set of incident-level columns. Using power query, I appended them into one table namely LFB_Incidents_All so that the analysis would cover the entire 2018-2024 span. I would then verify and correct the data types and make sure that dates, times, numeric measures and geographic fields were properly identified.

Missing values were pragmatically managed. There were no rows eliminated on key identifiers that were complete. IncidentGroup and other categorical nulls, as example IncidentGroup were recorded as Unknown, and gaps in numeric values were retained as null to prevent the averages being skewed. Since clear time-series analysis is better with a YearMonth column, the attendance time was changed to minutes. These measures narrowed down the research issue on the interaction between incident risk type, location and property risk and inequality in response times.

b) Use of Generative AI

I applied generative AI, as my supportive mechanism during the assessment. Initially, I requested a hand with the structure of the LFB datasets and a recommendation on how to unite the two periods in Power BI. The tool helped me to work with the tools of Power Query to combine the tables, verify data types and treat missing data in a manner that would not skew the analysis. It also proposed some handy calculated fields like a YearMonth String and response time in minutes.

I selected a six-visual, a KPI card, a line chart, a column chart, a map, a bar chart and a treemap and described the narrative value of each with the help of a generative AI which I used as a brainstorming partner when creating my dashboard. I made and optimized these images myself in Power BI according to underlying data. Another way I applied AI in enhancing the clarity and academic tone of my data story is to make all descriptions consistent with the visuals I created. Overall, generative AI helped me improve the structure and coherence of my work whereas the final choice of data preparation and visual design was my job.

c) What did you do well and what could you improve in your story and visualizations

I believe that the key strength of my story and visualisations is that they are closely associated with the research question. The six visuals serve a distinct purpose and are logically connected the KPI card sets the general workload; the time-series and grouped column chart investigate the distribution of the workload between the incident types, the map introduces geography, the response-time card assesses the work in the high-demand boroughs, and the treemap concentrates on property risk. This arrangement facilitates a viewer to track the storyline of the general background to specific understanding. The year and incident group add interactivity but does not overwhelm the user since the user can experiment with different perspectives and within the same story structure, which is provided using slicers.

The other strong point is the selection of easy to understand and familiar types of charts. The dashboard is easily interpreted by the majority audience using a line chart to show trends monthly and a column chart to show the composition of incidents. The map and treemap feature provide diversity because they distinguish the spatial and categorical patterns without introducing the unneeded complexity. I also concentrated on good labelling, the titles, axes, and legends of each visage are clearly spelled out. It was even better to transform the response time in minutes and generate a YearMonth field that enhanced readability and understanding by the user.

In several ways, it can be improved. Although the visuals are explicitly descriptive, the analysis is largely descriptive, and I do not test whether differences in boroughs or time trends are statistically significant. Further simplification of the insights with simple statistical summaries not involved in Power BI (confidence intervals, averages or correlations) is a potential addition to my future version of this. Furthermore, my dashboard is highly dependent on default Power BI colours; creating an active, narrative and easy to access colour scheme would increase readability and general communication.

To give more detailed geographical information, the map might be improved with filled map or filters based on the type of incident. The trees map is now saturated with different types of properties, using a top-N filter on frequent types and another visual on rare high-risk places would help to improve the message. Storytelling may be enhanced with short pieces of interpretive text or annotations on the dashboard that point out key peaks, high demand boroughs or predominant types of property to enable the viewer to understand what is being told by not necessarily reading an accompanying report.

References

- **Greater London Authority (2025) *London Datastore* [online]. Available at: https://data.london.gov.uk/ (Accessed: 29 December 2025).**

- **Greater London Authority (2025) *London Fire Brigade Incident Records*. London Datastore. Available at: https://data.london.gov.uk/dataset/london-fire-brigade-incident-records-em8xy/ (Accessed: 29 December 2025).**